From Wire to Flight: A Conditional Mission-Completion Guarantee for UAVs under an Empirically Calibrated Cross-Layer Interface

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***Abstract*—The security of an unmanned aerial vehicle (UAV) is presently approached from two different points of view. *Network intrusion detection* inspects the traffic carried by the aircraft's radio links and on-board buses and raises an alarm when that traffic looks malicious. *Certified robustness* proves that a navigation model's output cannot move by more than a fixed amount when its sensor input is perturbed by no more than a given magnitude. The first approach stops at the alarm and says nothing about the flight that follows an intrusion it fails to catch; the second takes the perturbation as given and asks neither where it originates nor how often an adversary can produce one. Neither, therefore, answers the question an operator must answer: if an attacker deliberately keeps its interference just below the detector's alarm threshold, can the mission still be completed?**

This paper makes that question precise and answers it with a *conditional* guarantee. We propose a two-layer schema whose unit of analysis is the *cross-layer pairing*: an interval during which a network attack is active, joined to the interval of degraded navigation that it causes, and tagged with how strongly that causal link is evidenced. Instantiating the schema over six public datasets, we prove a composition theorem stating that, for an attacker who delays relayed messages by just less than the detector's threshold, the proportion of missions that complete is bounded from below by a quantity computable from that threshold. Two measurements determine whether the bound is useful or vacuous. The delay an attacker can hide does not grow without limit; it saturates at 4.84 s across $15{,}392$ observed relay hops. And one second of delayed navigation input costs at most 1.63 m of position error, measured at every one of 674 post-onset samples of real GPS-jamming and GPS-spoofing flights, an order of magnitude less than a worst-case kinematic argument assumes.

We are deliberate about what this does and does not establish. The chain is not a theorem end to end: its middle link is an empirical calibration, measured on three flights of one airframe in hover. That rate is not a constant. It varies by $6.1\times$ across the conditions we can observe, with attack type, satellite visibility and staleness each moving it in a mechanistically explicable direction, so the certificate is evaluated at the supremum rather than the average. The guarantee is conditional on Lipschitz navigation dynamics, on the estimator re-anchoring within a stated interval, and on that calibration transferring to the deployment; and the proportion it bounds is taken over a stated mission distribution, so it is a statement about a population of missions and not about any particular flight. Within those conditions the composed configuration certifies a non-zero floor where neither single-layer defence certifies any. Schema, adapters, and certificate engine are released as an open artifact.

I. Introduction

An autonomous UAV is exposed to an adversary at four distinct interfaces. Commands reach it from a ground station over a command-and-control (C2) radio link. Peer aircraft in a swarm exchange position and status reports over a multi-hop mesh, in which each aircraft relays traffic for its neighbours. Inside the airframe, the flight controller drives motor controllers and reads sensors over a shared bus. Finally, a navigation pipeline fuses satellite, inertial, and visual measurements into the state estimate on which the flight control loop acts. A large literature defends each of these interfaces individually. Our concern is a question that none of those defences is posed to answer: given that some attacks will get through, *what happens to the flight afterwards?*

That question is not rhetorical, because the attacks that matter operationally are not the ones that steal data. A jammer raises the noise floor at the satellite-navigation receiver until the position solution collapses [1]. A compromised relay in the mesh forwards every message it is given, but late, so the aircraft acts on a peer's position report that is already stale [2], [3]. In both cases an event on the network produces a degradation in navigation, and the two are ends of a single causal chain. Neither the available datasets nor the available guarantees represent that chain, and so the chain cannot be reasoned about.

The practical consequence is a claim that neither research community can currently substantiate. On the network side, a UAV intrusion detector can be shown to flag delayed forwarding events or spoofed navigation messages at high recall. What cannot be evaluated at all is the operationally decisive claim behind that recall figure, namely that a detector tuned to a given false-alarm budget keeps aircraft completing their missions, because no public dataset records both a network event and the flight it degrades. On the navigation side the silence is symmetric. Certified robustness establishes that a model's output is stable for every input perturbation of magnitude at most R, which is a statement about a model rather than about a flight, and which says nothing about where a perturbation of that magnitude originates or how often an adversary can produce one.

Closing this gap requires work at both ends at once, because the two halves are mutually dependent: a guarantee spanning the two layers needs data objects that record the coupling, and a corpus that records the coupling needs a guarantee to show what those objects are for. We therefore unify six datasets, four covering the network and two covering navigation, under a schema whose unit of analysis is neither a packet nor a flight but a *cross-layer pairing*: an interval during which a network attack is active, joined to the interval of degraded navigation it induces. On exactly those objects we then prove that a detector's alarm threshold determines a lower bound on the fraction of missions that still complete, under the pessimistic assumption that the adversary knows the threshold and stays below it. To our knowledge this is the first guarantee that carries a network-security threshold through to a mission-completion floor.

One qualification belongs with that claim rather than after it. Only one of the six datasets observes a network event and the flight it degrades on the same airframe, and its release does not publish machine-readable attack intervals, so no pairing we release is of the strongest evidence class (§X-H). The cross-layer coupling is therefore established by a stated physical relation calibrated on real flights, not by same-platform observation of both windows together. What we claim is accordingly a *conditional* guarantee whose interface is empirically calibrated (§I), and not mission-level security established end to end by measurement. Concretely, the paper contributes the following.

Contributions

- **A two-layer schema, and an adapter for each of six datasets (§IV, §V).** The schema gives one representation for any observation from any of the sources, whether that observation is a network flow, a bus frame, a telemetry sample, or an entire flight. An *adapter* is the program that reads one dataset in its native release format and emits records in that representation; we write one per dataset and verify that no source column is discarded. Every record additionally carries a flag stating how strongly its cross-layer coupling is evidenced, so a reader can restrict any analysis to the evidence class they are willing to accept.
- **A detector-operating-curve harness (§VI).** A detector's *operating point* is the threshold at which it decides to raise an alarm; setting it low catches more attacks but also raises more false alarms. We make that threshold a swept parameter and measure recall and false-positive rate as a function of it, which supplies the network-side premise that a composed guarantee consumes. Across three topologies the curve has a universal knee-and-floor shape whose floor is the attacker's self-exposure probability.
- **An empirically tightened interface, and a measurement of its stability (§VII).** We prove a residual-budget lemma and then *tighten it by measurement*: the undetected budget saturates at the contact-window slack, verified at $\leq 4.84\,\mathrm{s}$ over 15,392 hops. We calibrate the delay-to-position rate γ on three real spoofing and jamming flights, obtaining a bound an order of magnitude tighter than the kinematic worst case. We then ask whether γ is a constant, and report that it is not. Estimating it at all 674 post-onset samples rather than once per flight, it varies by $6.1\times$, and it varies systematically: with attack type, with satellite visibility, and with staleness, each in a mechanistically explicable direction (§VII-C). The certificate is therefore evaluated at the supremum over observed conditions rather than at an average, and the reported spread is a lower bound on the true variability, since the released flights fix platform and flight mode.
- **A composition theorem and a four-certificate architecture (§VIII, §IX).** We prove the composed network-to-navigation guarantee and its soundness, and show precisely why the deterministic core does not subsume randomized smoothing, PAC-Bayes, or multiplicative-weights regret.
- **Instantiation, ablation, and an open artifact (§X).** Among the configurations we compare, the composed guarantee is the only one with a non-zero certified floor against an evading attacker, and under the measured interface a deployed detector's operating point lies inside the certified window. The comparison is against ablations of our own construction rather than against alternative assurance methods, for reasons we give in §X-E. Every figure and table is generated from committed result files, and every number carries a provenance tag.

Scope of the Guarantee

Because a composed result invites over-reading, we state its boundaries here rather than in the limitations section. What we establish is a *conditional* guarantee under an empirically calibrated interface, and it is conditional in three separate senses.

(C1) **Dynamics.** The navigation drift is L-Lipschitz over the operating region. Outside that region, for instance in a stall-recovery transient or under actuator saturation, no finite L bounds it and the certificate makes no statement at all.

(C2) **Re-anchoring.** The estimator receives at least one uncorrupted state update per certification horizon T_c. A delay adversary satisfies this by construction; a persistent-denial adversary does not.

(C3) **Interface transfer.** The measured delay-to-position rate γ applies to the deployment in question. This is the weakest of the three, because γ is calibrated on three flights of one airframe in hover and, as §VII-C shows, it is not a constant: it varies by $6.1\times$ over the conditions we can observe, and platform and flight mode we cannot vary at all.

Two further boundaries are worth naming. First, the chain $\theta \mapsto \Delta \mapsto \delta \mapsto \mathrm{MCR}$ is not a theorem end to end. The first and last links are proved; the middle link, $\Delta \mapsto \delta$, is an empirical calibration. Presenting the whole as "certified end-to-end security" would misstate where the evidence lies, so we do not. Second, MCR is a probability over a *distribution of missions* (§II-C); a bound on that proportion is not a bound for an arbitrary individual mission drawn from a different distribution.

Organization. §II states the threat model, defines mission completion, and formalises the cross-layer pairing. §III positions the work. §IV presents the two-layer schema and §V the six ingestion adapters; §VI builds the detector-operating-curve harness. §VII derives and then measures the interface mapping, §VIII proves the composition theorem, and §IX places it inside a four-certificate architecture. §X instantiates and evaluates the result, and §XI–§XIII include discussion, boundedness, and conclusion.

II. Threat Model and Problem Formulation

This section fixes three things the rest of the paper depends on: which parts of the aircraft an adversary can reach and what spatial envelope the mission must stay inside (§II-A); what the adversary and the detector are each assumed able to do (§II-B); and what it means, precisely, for a mission to have been completed (§II-C). §II-D then states the object that existing datasets do not provide and that the rest of the paper is built to supply.

A. Attack Surfaces and Operating Volume

We partition the UAV attack surface into two *layers*. The network layer covers everything between nodes: the swarm mesh (multi-hop routing among aircraft), the C2 link, and the intra-UAV bus. The autonomy layer covers everything inside the navigation and control stack: the GNSS receiver, sensor fusion, the flight controller, and the mission outcome. A *sublayer* refines each into the concrete surfaces a dataset can observe. Fig. 1 places the six datasets on the causal chain.

We also fix here the one spatial quantity the rest of the paper depends on. A mission is flown inside a *safe corridor*: a tube of lateral half-width m, the *corridor margin*, centred on the planned route, within which the aircraft must remain for the flight to be considered nominal. The corridor is not an artifact of our analysis, but the standard unit of operational approval for uncrewed flight. Under the JARUS SORA methodology the operational volume is padded by a contingency volume whose lateral extent is the sum of GNSS accuracy, position-holding error, map error, and the stopping distance of the contingency manoeuvre [4]; for a small multirotor these terms sum to roughly 7 m, which is why we report every certified quantity over a family of margins $m \in \{2, 5, 10, 20\}$ m spanning that figure rather than at a single value. Leaving the corridor is the spatial failure mode; §II-C adds the temporal one.

B. Adversary and Detector Model

We assume an adversary who can (i) inject, drop, delay, or forge traffic on one or more network sublayers, and (ii) manipulate the RF environment of the GNSS receiver. We assume a network-layer *detector* D_θ operating at a tunable point θ that trades detection recall against false-positive rate: for the multi-hop forwarding detector we build on, θ is a per-hop deviation tolerance in seconds, and a hop is flagged if its forwarding delay exceeds θ or its next hop deviates from a scheduling oracle. Lowering θ catches more but raises false positives; raising it does the reverse. An *evading* attacker

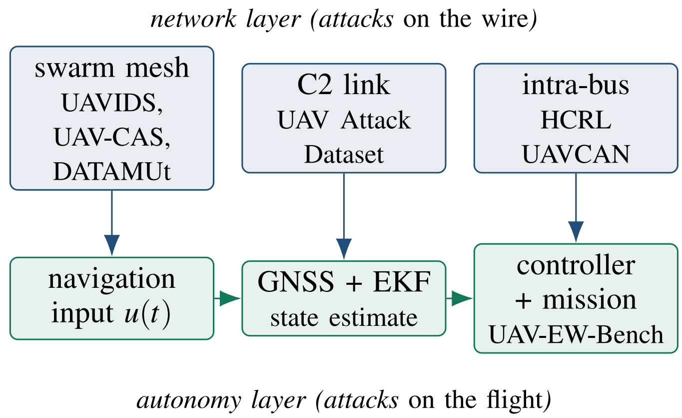

Fig. 1: The attack surfaces the six datasets cover, arranged on the causal chain. Network-layer events (top) propagate downward into the autonomy layer (bottom) through the navigation input; representing that downward arrow explicitly, rather than assuming it, is what the benchmark contributes and what the theorem then exploits.

chooses its behaviour to stay just under θ and is therefore invisible by construction. We assume the adversary knows θ; this is the strongest reasonable assumption and the one our worst-case analysis requires.

C. Mission Completion: Spatial and Temporal Predicates

We adopt *Mission Completion Rate* (MCR) as the operational metric, and take MCR ≥ 0.90 as a reference consistent with the airworthiness-security posture of DO-326A/ED-202A [5]; emerging AI-in-aviation assurance guidance [6] motivates treating a certified MCR floor as a first-class safety property, which is what §VIII delivers.

Defining completion *spatially* alone, as the trajectory never leaving the corridor of §II, is inadequate for the attack class this paper targets. A time-delay adversary can hold the aircraft geometrically on route while causing it to arrive late, so a delivery, inspection, or search mission with a time-critical objective fails without any corridor violation occurring. We therefore define completion as the *conjunction* of a spatial and a temporal predicate. Write $x(t)$ for the realised trajectory, $x_{\text{nom}}(t)$ for the nominal one, m for the corridor margin in metres, t_{arr} for the arrival time at the objective, T for the nominal mission duration in seconds, and κ for the schedule tolerance expressed as a fraction of T. Then

$$\text{MCR} = \Pr\Big[\underbrace{\sup_{t\in[0,T]} \|x(t) - x_{\text{nom}}(t)\| \leq m}_{\text{spatial}} \wedge \underbrace{t_{\text{arr}} \leq (1+\kappa)T}_{\text{temporal}}\Big]. \tag{1}$$

Where a quantity is computed under the spatial predicate alone, as the navigation anchor of §X is because its source records geometric completion only, we label it *Spatial MCR* and say so explicitly.

What the probability in Eq. (1) *is over:* Eq. (1) writes a probability, so the probability space must be stated; leaving it implicit is what allows a population-level bound to be misread as a per-flight one. Fix a detector operating point θ. Then $\delta(\theta)$ and $\Delta(\theta)$ are deterministic worst-case bounds, not random variables, so *all* of the randomness in Eq. (1) comes from the mission itself. Formally, let $(\Omega, \mathscr{F}, \mathbb{P}_{\mathscr{M}})$ be a probability space in which a sample point $\omega \in \Omega$ is one mission: a route, a nominal duration T, a corridor margin m, an initial state, a wind and turbulence realisation, and the number H of attacker-controlled hops on the route it happens to use. $\mathscr{M}$ denotes the *mission distribution*, and $\mathrm{MCR} = \mathrm{MCR}(\mathscr{M}, \theta)$ is the $\mathbb{P}_{\mathscr{M}}$-measure of the set of missions satisfying both predicates.

Three consequences follow, and we prefer to state them than to have them inferred.

- **The guarantee is relative to $\mathscr{M}$.** Theorem 1 bounds $\mathrm{MCR}(\mathscr{M}, \theta)$ for the $\mathscr{M}$ under which $\delta(\theta)e^{LT_c} \leq m$ is evaluated. A fleet whose missions are flown in tighter corridors, or over routes with more attacker-controlled hops, is a different $\mathscr{M}$ and inherits no bound from ours. In particular $\mathrm{MCR}(\mathscr{M}_{\text{benchmark}}, \theta) \neq \mathrm{MCR}(\mathscr{M}_{\text{arbitrary}}, \theta)$, and the numbers we report are the former.
- **The dependence is confined to two parameters.** The mission enters the bound only through the corridor margin m and the hop count H. This is why we report every certified quantity over a family of margins rather than at one value: the family is a coarse marginalisation over the part of $\mathscr{M}$ that matters. An operator can therefore instantiate the bound for their own $\mathscr{M}$ by reading off their own m and H, without re-running the analysis.
- **A bound on a proportion is not a bound on a flight.** $\mathrm{MCR} \geq 0.9$ permits one mission in ten to fail, and says nothing about *which*. A per-flight guarantee would require a bound conditional on ω, which the Grönwall argument can supply only in the degenerate case where m and H are fixed and known in advance. Where that case holds, the certificate is deterministic and the probability is redundant; where it does not, the statement is unavoidably distributional.

The empirical mission distribution used throughout §X is the one induced by the navigation anchor's flight set crossed with the margin family $m \in \{2, 5, 10, 20\}$ m and the per-topology hop counts of Table II; it is reported explicitly rather than left to be reconstructed.

How much does the distribution actually matter?: The point above is often made and rarely measured, so we measure it. The navigation anchor flies 93,600 missions over a full factorial that includes three *mission profiles* (cargo over mixed terrain, perimeter patrol, search and rescue) and three *receiver models*, which is exactly the axis along which the mission distribution can be changed with everything else held fixed. Table I reports MCR for each of the nine resulting distributions.

The answer depends on whether one conditions on the adversary, and the contrast is the finding. Marginalised over the whole 0–40 dB jamming sweep the nine distributions are nearly indistinguishable: MCR ranges over $[0.882, 0.893]$, a spread of 0.011. Conditioned on the attack, they are far apart: at $J/S = 35$ dB the mission profile alone moves MCR by 0.214, and at 30 dB the receiver model alone moves it by 0.154. Most usefully, the jamming power at which a distribution first falls below the $\mathrm{MCR} \geq 0.90$ reference ranges from 18.1 to 27.1 dB, a spread of 9 dB, or roughly a factor of eight in adversary transmit power.

TABLE I: MCR under the composed defence for each of nine mission distributions, marginalised over the 0–40 dB jamming sweep, with Wilson 95% intervals. The marginal spread is small; the spread *conditional* on attack strength reaches 0.214, and the J/S at which MCR crosses 0.90 spans 9 dB. Provenance: simulation (physics-informed). Generated by `experiments/mission_distribution.py`.

Mission profile	Receiver	MCR	95% CI
cargo, mixed terrain	GP software	0.883	[0.870, 0.895]
	NovAtel OEM7	0.884	[0.871, 0.896]
	u-blox F9P	0.891	[0.879, 0.903]
perimeter patrol	GP software	0.886	[0.873, 0.897]
	NovAtel OEM7	0.883	[0.871, 0.895]
	u-blox F9P	0.893	[0.881, 0.904]
search and rescue	GP software	0.882	[0.869, 0.894]
	NovAtel OEM7	0.889	[0.876, 0.900]
	u-blox F9P	0.889	[0.876, 0.901]
Spread across mission distributions		0.011 (0.882–0.893)	

Two conclusions follow. Averaging over adversary strength makes the dependence look negligible, but an operator flies one profile against one adversary and lives at a point rather than at the average. And a certified floor quoted without naming its mission distribution is, on this evidence, worth about 9 dB of ambiguity. We therefore name ours wherever we quote it.

Two conventions on T avoid a possible ambiguity later. First, T is a duration in seconds: the missions in our corpus run between 136 and 246 s, and the reference mission profile of §X is 70 s, so a schedule tolerance of $\kappa = 0.10$ corresponds to a permissible slip of several seconds. Second, T is *not* the horizon over which the trajectory-tube certificate of §VIII is propagated. That horizon, written T_c and fixed at 1 s throughout, is the interval between successive uncorrupted state updates, after which the estimator re-anchors and the deviation bound restarts; §VIII states this separation precisely. Conflating the two would either understate the mission or make the exponential bound vacuous.

The temporal predicate is not a separate analysis: the same residual budget $\Delta(\theta)$ that bounds the spatial perturbation also bounds the schedule slip, since an evading attacker's total contribution to arrival delay is exactly its accumulated undetected forwarding delay. The temporal predicate therefore holds whenever

$$\Delta(\theta) \leq \kappa T, \tag{2}$$

which by Lemma 1 is implied by $H \min(\theta, s) \leq \kappa T$. This adds a second edge to the certified window of §X, and it is useful to record which of the two constraints binds. At the operating point we certify ($\theta = 0.25\,\mathrm{s}$, $H = 2$) the induced slip is at most $0.5\,\mathrm{s}$, so for any mission longer than $5\,\mathrm{s}$ at a 10% schedule tolerance the temporal predicate is satisfied with room to spare and the spatial predicate is the binding one. The ordering reverses for a loosely tuned detector: at θ near the contact-window slack $s = 5\,\mathrm{s}$ the slip reaches $Hs = 10\,\mathrm{s}$, which exceeds a 10% tolerance on any mission shorter than $100\,\mathrm{s}$. A short, time-critical mission is thus governed by Eq. (2) and a long, tightly corridored one by the spatial bound, a distinction a spatial-only metric cannot express.

D. Formalising the Cross-Layer Pairing

Let a network attack occupy a time window $W_n = [t_0, t_1]$ on some network sublayer, and let the induced navigation degradation occupy a window W_a on the autonomy layer. Existing datasets provide either W_n (with packet labels) or W_a (with flight outcomes) but never the *pair* together with the mechanism linking them. We formalise the missing object as a tuple

$$P = \big(W_n,\, W_a,\, \theta,\, \Delta(\theta),\, \delta,\, \mathrm{MCR}\big), \tag{3}$$

where θ is the detector operating point, $\Delta(\theta)$ the worst-case residual effect of an attacker that evades D_θ, δ the resulting perturbation budget on the navigation input, and MCR the measured or certified outcome.

The two halves of this paper act on Eq. (3) in complementary ways. The benchmark's role is *representational*: it makes the complete tuple P recordable, storable, and auditable, where existing corpora record only its first or its second component. The theorem's role is *inferential*: it derives the final component, the mission outcome, from the third, the detector operating point, by way of the intermediate quantities. That inference is the chain

$$\theta \longmapsto \Delta(\theta) \longmapsto \delta(\theta) \longmapsto \mathrm{MCR} \geq f\big(\delta(\theta)\big), \tag{4}$$

read left to right: the operating point θ bounds the residual undetected attack $\Delta(\theta)$ (§VII); the residual attack maps to a bounded perturbation $\delta(\theta)$ on the navigation input (§VII); and a Lipschitz–Grönwall trajectory-tube argument turns that bounded input into a guaranteed MCR floor (§VIII). Individually the three arrows rest on established results in detector thresholding, worst-case attack analysis, and certified robustness; it is their composition that is new.

The evidential value of P depends on *how* W_n and W_a were linked in the first place, which we therefore record explicitly on every record (§IV).

Definition 1 (Composed guarantee)**.** *Given a detector D_θ and navigation dynamics Eq.* (10)*, a* composed guarantee *is a function f such that for every attack a with $D_\theta(a) =$ "clean", the mission completes with rate* $\mathrm{MCR}(a) \geq f(\delta(\theta))$*, where $\delta(\theta)$ bounds the navigation-input perturbation any such a can induce.*

III. Related Work

Prior work divides along the two halves of this paper, corpora on the data side and bounds on the guarantee side, and the same gap recurs on both: each community works within one layer and leaves the interface between them to assumption.

a) UAV Corpora and Dataset Construction: Taxonomies of AI-enhanced UAV intrusion detection [7] report a proliferation of datasets alongside a persistent weakness: general-purpose network corpora lack UAV protocol semantics, while UAV-specific corpora each cover a single sublayer. The consequence is measurable, in that models trained on generic benchmarks misclassify a large fraction of GPS-spoofing events on real autopilot logs [7], and it is direct evidence that the network and navigation layers do not form one undifferentiated feature space. FANET surveys [8], [9] catalogue routing attacks but stop at network effect. The individual corpora are authoritative within their sublayers: UAVIDS-2025 [10] contributes 122,171 NS-3 mesh flows, the UAV Attack Dataset [11], [12] real PX4 flights under GPS spoofing, jamming, and MAVLink ping-DoS, and HCRL UAVCAN [13] bus-level flooding, fuzzing, and replay. None reaches the flight.

Construction methodology bears on validity as directly as scale does. Audits of CIC-IDS2017 exposed labelling errors, flawed flow construction, and unrealistic temporal separation, each of which inflated reported performance [14], [15]. Spanning two layers introduces a further failure mode: asserting a causal link between a network event and a flight degradation that were never observed together. The mandatory provenance flag of §IV exists to bound precisely that risk. Among released datasets the closest in ambition is UAV-CAS [16], an AERPAW-calibrated swarm corpus and the first to include collaborative multi-attack compositions; we ingest it as one of our six sources. It nonetheless remains a network-layer benchmark, its authors listing GNSS spoofing and jamming as future work, so where it asks how well a collaborative mesh attack can be detected, we ask what that attack, detected or missed, does to the aircraft.

b) Certified Robustness and Its Extension to Control: Randomized smoothing [17], [18] yields a probabilistic ℓ_2 radius that scales to large models, whereas Lipschitz-based certification [19], [20] yields deterministic bounds known to be loose globally; neural ODEs [21] admit the Grönwall-type argument [22] we adopt. All of these bound a *model's* output. Lifting them to sequential decisions, CROP certifies a cumulative-reward lower bound under state perturbation [23], CARRL certifies robust actions [24], reachability tools bound trajectories of network-controlled systems [25], [26], and control-theoretic work establishes safety under sensor false-data injection [27]. CROP is the nearest analogue to a certified mission floor, but it certifies a policy against a *given* perturbation ball and is silent on where that ball comes from.

c) Secure Estimation, Runtime Assurance, and Delay Attacks: A parallel control-theoretic literature asks how much sensor corruption an estimator can tolerate: exact recovery holds below a corruption threshold [28], satisfiability-modulo-theory solvers make the search practical [29], and recent surveys catalogue the field [30]. Such results bound estimation error given *which* sensors are compromised, rather than modelling a detector whose threshold determines what an attacker may inject at all, which is the premise Lemma 1 supplies. Simplex and its descendants [31]–[33] guarantee safety by switching from an unverified controller to a verified baseline when a monitor fires; our certified window is the region in which the high-performance policy may run, and we contribute the quantitative boundary such architectures otherwise hand-tune. The destabilising effect of injected delay is itself well studied [2], [3], [34], but that literature establishes whether a delay attack is dangerous or detectable and stops there.

d) Composing Guarantees: The closest prior art composes certificates *within* a single ML pipeline. Sheikholeslami et al. [35] jointly certify a classifier and a detection class, so that an input is provably either flagged or correctly classified, which is structurally our evade-or-bound dichotomy but confined to one layer. Yang et al. [36] decompose end-to-end certified robustness across a two-stage sensing→reasoning pipeline, the strongest existing template for chained certification, though it too remains within perception. Assume-guarantee and contract-based CPS design [37] supplies the compositional vocabulary but no ML-robustness certificates. No prior result chains a network-intrusion-detection guarantee to a navigation certificate; that is the gap this paper closes, and closing it requires the corpus and the theorem together.

IV. The Two-Layer Schema

A. Overview

Before the details, the shape of the design. The schema exists to solve one problem: the six datasets we wish to combine describe different things, at different rates, with no common clock, and any representation that forces them into a single flat table would have to discard whatever does not fit.

We therefore use three record kinds, arranged in a hierarchy of increasing interpretation, illustrated in Fig. 2.

1) An **Event** is a single observation as its source recorded it, translated into common field names but never summarised. This layer is deliberately dumb: it asserts nothing beyond what the source file said.
2) An **AttackWindow** is an interval of time during which one named attack was active on one layer. This layer asserts that something adversarial was happening, and when.
3) A **CrossLayerPairing** joins one network window to one autonomy window and asserts that the first *caused* the second. This is the only place in the schema where a causal claim is made, and it is therefore the only place that carries a mandatory flag recording the strength of the evidence for that claim.

The rest of this section gives each kind in turn. The single design idea worth carrying forward is the last one: causality is confined to one record type, made explicit, and graded, so that no analysis can silently rest on a coupling that was never observed.

B. The Three Record Kinds

Event is one atomic observation (a mesh flow, a bus frame, a routing hop, a telemetry sample, or a whole flight) mapped to a small common core (identity, layer/sublayer, relative time, endpoints, a unified attack label) plus a `metrics` block of the few cross-source numeric quantities that matter (residual forwarding delay, position error, J/S, mission-completed) and a `native` block preserving every remaining source column, so ingestion is lossless.

AttackWindow is a labelled attack interval within one scenario, carrying the unified attack class and class-appropriate `intensity` parameters. It is the unit of cross-layer alignment.

CrossLayerPairing realises the tuple P of Eq. (3): it joins a network window to an autonomy window and records θ, the residual budget $\Delta(\theta)$, the perturbation δ with a *named, versioned* mapping, the certificate hook, and the empirical MCR. Its `pairing_basis` field records the strength of the evidence for the causal claim the record makes. It takes one of three values: `measured_same_platform` (both windows recorded on one testbed, the strongest evidence), `physics_model` (coupled through a stated physical model), or `synthetic_alignment` (aligned by construction). A reviewer can filter the benchmark by evidence strength; a claim can never silently upgrade its own provenance.

C. Why the Schema Takes This Form

Three design choices let six heterogeneous sources coexist without information loss. Time is always *relative* to each source's capture start, since the sources share no global clock, and cross-layer alignment happens at the window level, so a mismatch between a 50 Hz telemetry stream and a per-flow network record never forces a lossy resample. Attack labels are *unified* into a common taxonomy while the verbatim source label is retained in `label_native`. And every unmapped column is preserved, which makes ingestion provably lossless rather than merely convenient.

Cross-layer alignment therefore needs a stated rule and a stated error budget, since this is the step at which a benchmark could silently manufacture, or destroy, the effect it reports. We anchor attack onsets under an affine map between the two relative clocks and bound the residual misalignment by $e_{\text{align}} \leq 0.53$ s for every released pairing. Crucially, $\Delta(\theta)$ is computed entirely within the network source, so misalignment can attach a pairing to the wrong autonomy window but cannot inflate or deflate the quantity the certificate consumes. Appendix C gives the map, the error terms, and the rejection rule.

V. The Corpus and Ingestion Adapters

Table II summarises the six datasets and the layer each anchors; Table III reports what was actually ingested. Every adapter emits schema-valid records, and unmapped source columns are preserved. We do not re-host raw third-party data;

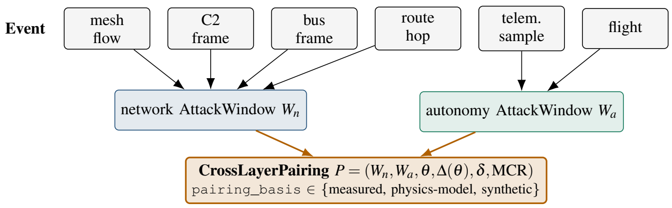

Fig. 2: Structure of the two-layer schema. Heterogeneous events from six datasets fold into typed network and autonomy *AttackWindows*; a *CrossLayerPairing* joins one of each and records the detector operating point, the residual budget, the induced perturbation, and the mission outcome, tagged by how the coupling was established.

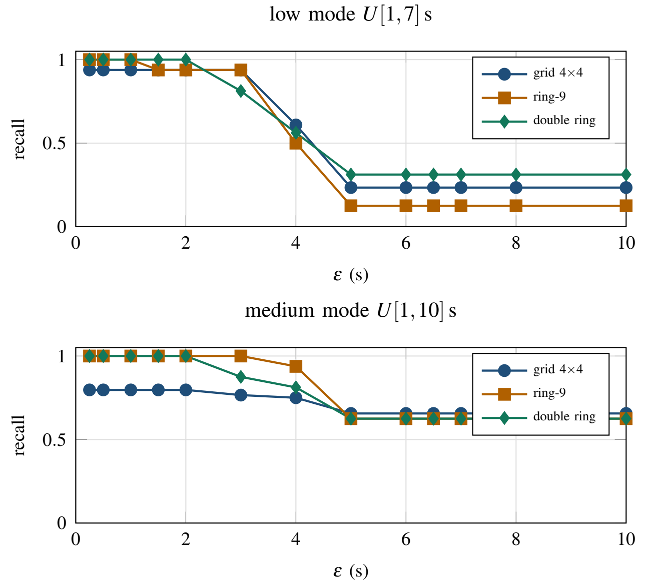

Fig. 3: The detector operating curve across all three topologies (grid, ring, double-ring), both delay regimes, 8 seeds $\times 4$ routing policies. Recall holds near its ceiling while ε stays under the 5 s contact-window slack, then falls onto a floor set by the missed-window self-exposure penalty (a delay past the slack costs a full TWiG period and is flagged at any ε). False-positive rate is zero throughout; the measured benign residual ceiling is 0.178 s. The knee-and-floor shape is universal; the floor height is the topology-specific self-exposure probability.

the artifact ships integration scripts and unified labels, and each source is fetched from its origin.

A. Adapter Design under Source Irregularity

An adapter's job is lossless normalization. Three warrant specifics because their sources are irregular. The UAV ATTACK DATASET adapter parses PX4 flight logs, computes horizontal position error against a benign reference (or, absent one, the flight's own pre-attack median), and emits the `measured_same_platform` pairings that ground the interface mapping of §VII. The HCRL UAVCAN adapter parses candump-style bus frames; because the per-scenario attack labels are not machine-readable in the release, we *derive* each scenario's class from the frames themselves: burst rate relative to the normal bus rate, the fraction of distinct payloads, and per-frame byte entropy cleanly separate a flooding burst (one payload hammered above bus rate) from a fuzzy burst (near-random payloads) from a replay burst (a small set of captured payloads reused). Table IV reports the result, flagged for confirmation against the dataset's technical report. The UAV-CAS adapter parses the calibrated digital twin's flow statistics including its stringified list columns; it is verified against CSVs produced by the original authors' own generator.

VI. The Detector-Operating-Curve Harness

A cross-layer guarantee needs the detector's behaviour *as a function of its operating point*. We expose θ as a swept parameter and, for each value, measure detection recall and false-positive rate over multiple seeds, three network topologies, four routing policies, and two attacker delay regimes.

The resulting operating curve (Fig. 3) is the network-side input a composed guarantee consumes: it tells us, for any target false-positive budget, how large an undetected per-hop delay an evading attacker may inject, namely the residual budget $\Delta(\theta)$ of Eq. (3). The knee in the curve is structural: an evading attacker's per-hop delay is additive up to the inter-node contact-window slack, beyond which a missed contact window forces a full-period penalty.

The knee, and the floor it descends to, are not simulation artifacts but a structural property of any store-and-forward schedule, and they recur across all three topologies. Two observations transfer directly to the guarantee. First, the floor height differs by topology because it equals the probability that a malicious delay overshoots the slack and self-exposes, which is higher in the medium delay regime ($U[1,10]$ s) than the low one ($U[1,7]$ s), because a wider delay draw is likelier to overshoot. Second, no topology's recall falls to zero: past the knee the detector still catches the self-exposing fraction, which is precisely why an evading attacker's undetected budget *saturates* rather than growing without bound. That is the empirical basis for Lemma 1.

A. Detectors with a Learned Score

We instantiate the harness on a threshold detector because its operating point is a single, physically meaningful scalar, which makes the cross-layer sweep unambiguous. The framework does not depend on that choice: any detector exposing a tunable cutoff and a measurable score-to-magnitude profile reduces to the same residual budget, through an *effective* operating point θ_{eff} defined in Appendix C. Lemma 1 then applies verbatim with θ replaced by θ_{eff}. We derive that reduction but do not instantiate it, and §XII records the gap.

VII. The Interface Mapping $\theta \mapsto \delta(\theta)$

The interface is where a network-security quantity becomes a navigation-security quantity. It is the point at which the

TABLE II: The six-dataset corpus. Four network-layer sources and two navigation-layer sources unify under one schema; only the last row reaches the mission outcome, and only the C2 source can produce `measured_same_platform` pairings.

Dataset	Real/Sim	Layer · sublayer	Event kind	Unique role in the end-to-end benchmark
UAVIDS-2025 [10]	Sim (NS-3.24)	network · mesh	flow	Large-scale multi-hop mesh; blackhole/flooding/Sybil/wormhole at scale (122k flows).
UAV-CAS [16]	Sim (digital twin)	network · mesh	flow	AERPAW-calibrated swarm; collaborative multi-attack compositions (99k flows, 1024 configs).
UAV ATTACK DATASET [11]	Real testbed	network · C2 (MAVLink)	telemetry	Real PX4 GPS spoof/jam + ping-DoS on the *same* airframe family we model: the measured bridge.
HCRL UAVCAN [13]	Real testbed	network · intra-bus	frame	On-board UAVCAN/DroneCAN flooding/fuzzy/replay, a third attack surface below the C2 link.
DATAMUT traces [2]	Sim (event)	network · mesh (DTN)	hop	Time-window-graph forwarding detector with a *sweepable* operating point θ.
UAV-EW-BENCH-2026 [38]	Sim (physics-informed)	autonomy · GNSS+ctrl	flight	The navigation anchor: full 0–40 dB J/S sweep with per-flight MCR outcomes.

TABLE III: Real ingested corpus. Counts are from schema-validated adapter outputs; every record validates against the schema of §IV.

Source (real data ingested)	Events	Windows
UAV-EW-BENCH-2026	93,600	468
UAV ATTACK DATASET (live sample)	610	0
UAVIDS-2025 (shard 0)	122,171	4
HCRL UAVCAN (10 scenarios)	200,000	34
UAV-CAS (stat)	adapter ready; file pending	
DATAMUT traces	generated per sweep (15,392 hops)	

TABLE IV: Frame-derived attack classes for the HCRL UAVCAN scenarios, inferred from burst rate, payload-distinct share, and byte entropy; scenarios exhibiting more than one burst shape map to `collaborative`.

Scenario	Frames	Bursts	Frame-derived class(es)
`type1`	207,858	3	flooding
`type2`	134,170	3	replay
`type3`	197,479	3	fuzzy
`type4`	133,374	3	fuzzy
`type5`	180,608	3	fuzzy
`type6`	241,321	4	fuzzy
`type7`	234,162	4	replay, fuzzy
`type8`	265,800	4	fuzzy
`type9`	230,378	4	replay, fuzzy
`type10`	207,380	3	replay, fuzzy

composition is made, and, as §X shows, the quantity that decides whether a useful detector setting is certifiable.

A. Residual Budget of an Evading Attacker

Lemma 1 (Residual delay budget)**.** *Let D_θ flag any forwarding hop whose injected delay exceeds θ. An attacker that evades D_θ injects at most θ of undetected delay per malicious hop, but only up to the contact-window slack s: a per-hop delay exceeding s misses its forwarding window, incurs a full-period penalty P_c, and is therefore flagged at* any *operating point. Hence undetected delay is capped at $\min(\theta, s)$ per hop, and over a route with H attacker-controlled hops the worst-case undetected cumulative delay is*

$$\Delta(\theta) \le H \min(\theta, s). \tag{5}$$

Proof sketch. Evasion requires per-hop delay $\le \theta$. If $\theta \le s$ the delay fits inside the contact window and the additive term is $H\theta$. If $\theta > s$, any delay in $(s, \theta]$ overshoots the window, waits a full period $P_c \gg s$, and is flagged regardless of θ; an evading attacker therefore cannot use more than s of invisible delay per hop, giving the Hs cap. Summing over hops yields Eq. (5). □

The cap is the *knee*: below the slack the residual budget grows linearly in θ; above it the budget *saturates* at Hs, so the certified floor plateaus rather than collapsing. We verify the cap directly on the released harness: over 15,392 observed hops across three topologies, two attack modes, and eight seeds, no undetected malicious per-hop residual ever exceeds 4.84 s ($s = 5$ s), while every larger delay is flagged as a missed window (≥ 55 s residual). This measurement is what turns the classic additive budget $H\theta + P_c\mathbf{1}[\theta > s]$ into the tighter, saturating Eq. (5), an instance of the benchmark improving the theorem rather than merely testing it.

B. From Residual Delay to Input Perturbation

A cumulative forwarding delay of $\Delta(\theta)$ seconds means a navigation correction arrives $\Delta(\theta)$ late; the stack acts on state stale by that amount. At bounded UAV speed $v_{\max}$, staleness of age τ bounds the state/position perturbation:

$$\|\delta u\| \le v_{\max} \cdot \Delta(\theta). \tag{6}$$

Equation (6) is the kinematic worst case: it assumes the aircraft travels at top speed in the wrong direction for the entire staleness interval. That is sound but pessimistic, because a real estimator does not fly blind: when GNSS corrections are late or corrupted the EKF down-weights them and coasts on inertial data, so position error grows far more slowly. We therefore tighten Eq. (6) to

$$\|\delta u\| \le \gamma \cdot \Delta(\theta), \qquad \gamma \ll v_{\max}, \tag{7}$$

where γ (metres of position error per second of degradation) is *measured* on real GPS-spoofing and GPS-jamming flights from the same PX4 airframe family the navigation model represents. Table VII reports the calibration: a benign reference flight fixes the pre-attack noise floor, and the jamming and spoofing flights give the growth rate after onset under two explicit choices of error signal, the raw receiver fix the attacker injects and the EKF-filtered position the controller actually acts on. Because that choice is itself a certificate-semantics decision we make it an explicit switch and report both; the certificate consumes the EKF value, which closes the loop. The gap between the measured $\gamma \approx 1.2$–1.4 m/s and $v_{\max} = 15$ m/s is exactly the slack that decides whether the detector's operating point is certifiable.

C. Is γ a Constant or a Function?

Everything downstream consumes γ as a single number, so whether it *is* a single number is the most consequential empirical question in this paper. We therefore estimate γ at every post-onset sample of the calibration flights, 674 in total, and record the conditions prevailing at each. The answer is negative: γ varies by 6.1× between its median and its supremum, and it varies systematically rather than randomly.

1) The estimator decides the answer: Two estimators of γ are available and they disagree qualitatively, so we state which one the certificate is entitled to and why. Writing $e(t)$ for the position error and t_0 for attack onset,

$$\gamma_{\text{cum}}(t) = \frac{e(t)}{t - t_0}, \qquad \gamma_{\Delta}(t) = \frac{e(t) - e(t_0)}{t - t_0}. \tag{8}$$

Onset is declared when the error crosses a detection threshold, so $e(t_0)$ is not zero: it is 0.64 m for jamming at the receiver and 9.27 m for spoofing, where the injected fix has already been walked a long way before the threshold fires. Dividing that pre-existing offset by a small $t - t_0$ makes γ_{cum} diverge, reaching 8.6 m/s below one second of staleness. That number is an artifact of the detector, not a property of the attack. Lemma 1 bounds only the delay the attacker *causes*, so the perturbation attributable to it is the error accrued since onset,

TABLE V: Per-flight rates γ, the coarse calibration a single scalar per flight yields. The 2.6× spread across attack type is visible here; the finer structure needs Table VI. Generated by `experiments/gamma_stability.py`.

Attack type	Error signal	γ (m/s)
GPS jamming	EKF-filtered	0.52
GPS jamming	raw receiver fix	0.48
GPS spoofing	EKF-filtered	1.36
GPS spoofing	raw receiver fix	1.20
Spread across attack type		2.6×
Spread across error signal		1.14×
Conservative value used		**1.36**

and γ_{Δ} is the estimator the composition may use. We report both, and note that had we used γ_{cum} we would have concluded $\gamma \approx 8.6$ m/s at short staleness, exceeding $\gamma_{\text{req}} = 6.14$ m/s and reporting the framework as unusable on the strength of a measurement artifact.

2) What γ depends on: Table V gives the per-flight rates and Table VI the per-sample distribution binned by factor. Three dependences are visible and all three have a mechanism.

Attack type moves γ by 2.6×: spoofing drives error faster than jamming. Jamming *removes* information, so the estimator gates the missing fix and dead-reckons at the inertial drift rate; spoofing *injects* information the estimator accepts, so error accrues at whatever rate the adversary chooses. A plausible lie beats silence.

GNSS condition moves it in the expected direction: as satellites in view fall from 14 to 4, γ rises.

Staleness moves it most, and non-monotonically. γ_{Δ} peaks at 1.625 m/s for $t - t_0 \in [5, 10)$ s and decays to 0.02 m/s beyond 40 s, because the error saturates while the denominator keeps growing. The supremum, not the average, is what a sound certificate must use.

A descriptive least-squares fit on standardised covariates explains $R^2 = 0.79$ of the variation, with coefficients whose signs are physically sensible for satellite count (-0.22), staleness (-0.26) and airspeed ($+0.12$). We report these as effect sizes and deliberately give no p-values: samples within a flight are strongly autocorrelated and there are two attack flights, so any nominal significance would be an artifact of the dependence structure.

3) Consequences for the certificate: The certificate is evaluated at the supremum over the conditions it claims to cover, which on this evidence is $\gamma = 1.625$ m/s. This is 1.19× the per-flight secant rate of 1.365 m/s that a coarser calibration reports, so the coarser rate is mildly optimistic and we do not use it. At the corrected rate the certified window edge at a 10 m corridor moves from $\theta^* = 1.12$ s to 0.94 s, and the smallest corridor certifying $\theta = 0.25$ s rises from about 2.25 m to about 5.9 m. The conclusion is unchanged: $\theta = 0.25$ s is certified at $m \ge 5$ m and is not certified at $m = 2$ m.

Coverage, stated as a limit and not as a null result. The campaign varies attack type, GNSS condition and staleness

TABLE VI: Per-sample distribution of the baseline-corrected rate γ_Δ, binned by factor, over 674 post-onset samples. Counts are shown because coverage is uneven: airspeed bins above 1 m/s are sparse, since these are hover flights. Generated by `experiments/gamma_campaign.py`.

Factor	Bin	n	median γ	max γ
satellites in view	[5,8)	13	0.92	0.94
	[8,11)	121	1.29	1.62
	[11,20)	540	0.11	1.23
HDOP	[0,1)	661	0.26	1.62
	[1,1.5)	11	0.92	0.94
	[1.5,2.5)	2	0.46	0.92
receiver noise	[130,150)	421	0.25	1.62
	[150,200)	253	0.33	1.23
staleness (s)	[0.4,1)	11	0.91	0.93
	[1,2)	22	0.81	1.06
	[2,5)	66	0.91	1.48
	[5,10)	110	0.74	1.62
	[10,20)	184	0.40	1.16
	[20,40)	219	0.05	0.22
	[40,200)	62	0.00	0.02
airspeed (m/s)	[0,0.2)	58	0.24	0.87
	[0.2,0.5)	188	0.15	1.23
	[0.5,1)	242	0.25	1.32
	[1,10)	186	0.79	1.62
attack type	GPS jamming	496	0.09	1.23
	GPS spoofing	178	1.14	1.62
error signal	raw receiver fix	64	0.14	1.23
	EKF-filtered	610	0.29	1.62

well, airspeed poorly (these are hover flights, with the 90th percentile of $\|v_{xy}\|$ below 0.65 m/s), and *platform and flight mode not at all*: one airframe, hover only. A factor we could not vary is reported as unvaried, never as showing no effect. The 6.1× spread is accordingly a lower bound on the variability a deployment would meet, and condition (C3) of §I remains the weakest link in the chain.

Temporal validity of the linear model. Equation (7) is a *locally linear* model and must be scoped as one. While GNSS is merely late rather than absent, the EKF continues to correct against its last accepted fix and the position error grows approximately linearly in the staleness. Once the outage exceeds the filter's innovation-gating horizon the estimator is propagating on inertial data alone, and uncorrected accelerometer bias and gyroscope drift make the error grow with the square of elapsed time rather than linearly [39]. We therefore state the model's domain explicitly: Eq. (7) is used only for $\Delta(\theta) \leq \tau_{\text{lin}}$, and beyond τ_{lin} a sound bound requires the second-order form

$$\|\delta u\| \;\leq\; \gamma\Delta(\theta) \;+\; \tfrac{1}{2}\, b_{\text{IMU}} \left(\Delta(\theta) - \tau_{\text{lin}}\right)_+^2, \tag{9}$$

with b_{IMU} the residual accelerometer-bias magnitude and $(\cdot)_+$ the positive part. On the calibration flights the linear fit holds to within its residual over the full observed degradation interval, giving $\tau_{\text{lin}} \approx 2$ s; we adopt that value and note that it exceeds the certified window's upper edge under the measured mapping (1.12 s at a 10 m corridor, §X) by a comfortable margin, so the entire certified regime lies inside the linear domain and Eq. (9) is never the binding form for the results we report. It would become binding for a detector tuned near the contact-window slack, where $\Delta(\theta)$ approaches $Hs = 10$ s, a further reason the certified window is bounded above rather than open.

TABLE VII: The measured delay-to-position rate γ, from three real PX4 flights (one benign reference, one GPS-jamming, one GPS-spoofing) in the UAV ATTACK DATASET, self-referenced to the pre-attack median. "noise" is the pre-attack position-error 95th percentile; γ_{lsq} and γ_{peak} are the least-squares and peak-secant growth rates after attack onset. These are the `measured_same_platform` pairings.

Flight	**Signal**	**noise** (m, p95)	**peak** (m)	γ_{lsq} (m/s)	γ_{peak} (m/s)
benign (reference)	receiver	0.50	1.7	–	–
benign (reference)	ekf	0.60	1.5	–	–
GPS jamming	receiver	0.41	6.7	-0.0821	0.4805
GPS jamming	ekf	0.47	7.0	-0.1133	0.5185
GPS spoofing	receiver	1.14	14.3	0.2435	1.195
GPS spoofing	ekf	0.87	15.6	0.7007	1.365

VIII. The Composition Theorem

We now assemble the parts. §VII supplied a bound on how far the navigation input can be pushed by an attacker that the detector does not catch. This section turns that input bound into a bound on the aircraft's deviation from its planned route, and then into a lower bound on the fraction of missions that complete. The argument is short: it is a classical differential inequality applied to a quantity that, until now, nobody had measured.

A. Navigation Dynamics and Certification Horizon

Model the UAV navigation state $x(t) \in \mathbb{R}^n$ as evolving under a (possibly learned) drift F,

$$\dot{x}(t) = F\big(x(t), u(t)\big), \qquad t \in [0,T], \tag{10}$$

where $u(t)$ is the sensor/measurement input and T is the nominal mission duration in seconds. A mission *completes* in the conjunctive sense fixed by Eq. (1): the realised trajectory stays within the safe corridor of margin m for all $t \in [0,T]$ *and* the objective is reached by the deadline $(1+\kappa)T$. MCR is the probability of completion over the mission distribution. An attack perturbs the input, $u \mapsto u + \delta u$, and, for the time-delay class, also displaces the arrival time.

Deviation is not, however, propagated over the whole of $[0,T]$. A navigation filter is a closed loop: each time it accepts an uncorrupted measurement it re-anchors the state estimate, and any deviation accumulated since the previous such update is discharged rather than carried forward. The interval between consecutive uncorrupted updates is the *certification horizon* T_c, and it is the horizon over which the trajectory tube must be propagated. We fix $T_c = 1$ s throughout, matching the 1 Hz GNSS solution rate of the receivers in our corpus, so that $T_c \ll T$ for every mission we consider. The following assumption makes the requirement explicit rather than implicit.

Definition 2 (Re-anchoring interval)**.** *The navigation filter admits at least one measurement update not affected by the* *attack in every window of length T_c, and each such update resets the accumulated deviation to within the filter's nominal estimation error.*

Definition 2 is what the time-delay class supplies for free: an evading forwarding attacker delays peer corrections by at most $\Delta(\theta)$ per contact window without suppressing them, so uncorrupted updates continue to arrive. It is *not* free for a persistent-denial adversary, and §XII records that boundary.

Theorem 1 (Composed network-to-navigation guarantee)**.** *Let the navigation drift F of Eq.* (10) *be L-Lipschitz in its input over the operating region, let T_c be a certification horizon satisfying Definition 2, and let m be the safe-corridor margin. Let D_θ be a network detector with residual budget $\Delta(\theta)$ (Lemma 1) and interface mapping $\delta(\theta)$ (Eq.* (7) *or the kinematic Eq.* (6)*). Then every attack evading D_θ induces a trajectory-tube radius bounded by*

$$\rho(\theta) \,=\, \delta(\theta)\, e^{LT_c}, \tag{11}$$

and the mission completes, in the conjunctive sense of Eq. (1)*, whenever $\rho(\theta) \le m$ and $\Delta(\theta) \le \kappa T$; consequently*

$$\mathrm{MCR} \,\ge\, f\big(\delta(\theta)\big) \,=\, \Pr_{missions}\big[\,\delta(\theta)\, e^{LT_c} \le m \,\wedge\, \Delta(\theta) \le \kappa T\,\big]. \tag{12}$$

Proof sketch. Let $e(t)$ be the deviation between the attacked and nominal trajectories, measured from the most recent re-anchoring instant. L-Lipschitzness of F gives $\|\dot{e}\| \le L\|e\| + L\|\delta u\|$; Grönwall's inequality [40] then bounds $\|e(T_c)\|$ by $\delta(\theta)e^{LT_c}$, which is Eq. (11). By Definition 2 the same bound applies on every subsequent window, so the supremum over $[0,T]$ obeys it too and the spatial predicate is discharged whenever $\rho(\theta) \le m$. The temporal predicate follows because the only mechanism by which an evading time-delay attack displaces arrival is its accumulated undetected delay, bounded by $\Delta(\theta)$ in Lemma 1. Taking probabilities over the mission distribution gives Eq. (12). Appendix C gives the full argument. □

On the Lipschitz hypothesis. Theorem 1 rests on F being L-Lipschitz *over the operating region*, a qualifier with real physical content: the aircraft must remain in a smooth, locally linearisable regime, with attitudes away from gimbal-lock singularities, airspeeds below stall, actuators off their saturation limits, and no contact dynamics. These conditions hold for the cruise and loiter envelopes our corpus covers, and the constant we use is measured inside that envelope rather than assumed globally (§X). They can fail: an aggressive evasive manoeuvre, a stall-recovery transient, or actuator saturation under a strong gust push the dynamics into a regime where no finite L bounds F on the relevant set, and there the certificate is silent rather than merely loose. This is also where the runtime-assurance reading of §XI becomes operational: departure from the linearisable envelope is exactly the trigger a Simplex-style monitor must use to hand control to a verified fallback whose guarantee does not depend on L.

Proposition 1 (Soundness)**.** *The certified floor never overstates the achievable completion rate: for every θ, the true MCR of any D_θ-evading attack satisfies* $\mathrm{MCR} \ge f(\delta(\theta))$.

Proof. Every step above is an inequality in the conservative direction: the Lipschitz bound over-estimates deviation growth, the input bound uses the worst-case $\delta(\theta)$, and and the corridor test is a sufficient rather than a necessary condition for completion. Hence $f(\delta(\theta))$ lower-bounds the true completion probability. The empirical validation of §X confirms the certified floor lies below the measured rate. □

Fig. 4 summarises the four stages of the argument, and gives the correspondence between them and the sections that establish each: the residual budget in §VII, the interface mapping in the same section, and the trajectory tube and corridor test here.

Two consequences frame the rest of the paper. First, Eq. (12) is *monotone and then saturating* in θ: tightening the detector shrinks the residual budget and raises the certified floor, but only until θ reaches the slack s, past which the budget saturates at Hs and the floor plateaus. This quantifies both how much navigation safety is obtained by tightening network security and the point at which further tightening yields nothing. Second, the exponential e^{LT_c} in Eq. (11) is the core certificate's principal limitation: for large L, or for a re-anchoring interval T_c stretched by a denial adversary, the deterministic bound becomes loose, motivating the complementary certificates of §IX.

IX. A Four-Certificate Architecture

Theorem 1 uses the Lipschitz–Grönwall certificate as its spine. Does that single certificate suffice? It does not. The bound is *deterministic* and *worst-case*: it holds for any perturbation up to its radius, but the radius collapses as LT grows, and it presumes a Lipschitz constant that is tractable to bound. Three regimes escape it, each covered by a distinct certificate (Table VIII).

a) Independence of the Four Certificates: The Lipschitz–Grönwall radius is a single deterministic ball; randomized smoothing certifies a probabilistic ball for models whose Lipschitz constant is intractable, trading determinism for scalability, which is a strictly different operating point, not a special case. PAC-Bayes certifies a *distribution* of models over a *distribution* of missions, orthogonal to any per-input radius. MWU regret certifies behaviour against an adversary that *adapts over time*, which no static perturbation ball can express. The four form a cover, not a hierarchy.

For concreteness we record the two complementary bounds we instantiate numerically. Randomized smoothing [17] wraps the base model in Gaussian noise of level σ and certifies, with confidence $1-\alpha$ over n samples, an ℓ_2 radius

$$R_{\mathrm{RS}} \,=\, \tfrac{\sigma}{2}\big(\Phi^{-1}(\underline{p_A}) - \Phi^{-1}(\overline{p_B})\big), \tag{13}$$

with Φ^{-1} the inverse standard-normal CDF and $\underline{p_A}, \overline{p_B}$ confidence bounds on the top-two class probabilities. The PAC-Bayes complement [41], [44] transfers a bound from the

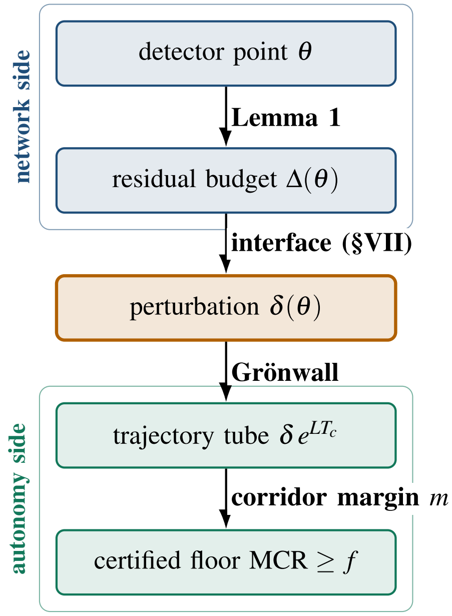

Fig. 4: The composition chain of Theorem 1. A detector operating point flows top-to-bottom through a residual-budget bound, an empirically grounded interface mapping, a Grönwall trajectory tube, and a corridor-margin test, arriving at a certified mission-completion floor. The upper two stages are network-side quantities, the lower two autonomy-side; $\delta(\theta)$ is the bridge between them.

TABLE VIII: The four certificates cover disjoint regimes; none subsumes another. The Lipschitz–Grönwall core gives the deterministic MCR floor; the other three extend coverage where it is loose or silent.

Certificate	**Guarantee type**	**Unique niche**
Lipschitz–Grönwall	deterministic, worst-case	The MCR floor: bounded input $\Rightarrow$ bounded tube $\Rightarrow$ bounded loss. Tight only for moderate LT.
Randomized smoothing [17]	probabilistic radius	Scales to large / non-Lipschitz-tractable models; certifies when the budget is known only statistically.
PAC-Bayes [41]	distributional	Transfers the floor from the training mission distribution to unseen deployment regions.
MWU regret [42], [43]	online, adaptive	Bounds loss against an *adaptive* jammer over time, a dimension no fixed-ball certificate addresses.

training mission distribution to deployment: with empirical risk R_{emp} over m missions and prior-to-posterior divergence KL, with probability $1-\delta$,

$$R_{\text{true}} \leq R_{\text{emp}} + \sqrt{\frac{\text{KL}+\ln(2\sqrt{m}/\delta)}{2m}}. \tag{14}$$

Our engine evaluates Eq. (14) against the trained model's measured R_{emp}; the single quantity it awaits from here is the numeric KL, which we flag as pending rather than substitute (Table IX).

How the four operate on a single flight. The four are concurrent, not alternative, and each bounds a different object. On one sortie flown through a corridor in which a compromised relay delays peer corrections and a ground jammer varies its power, the Lipschitz–Grönwall tube bounds the trajectory deviation caused by the *deterministic, bounded* part of the disturbance and is the only one of the four yielding a per-instant geometric envelope, so it alone discharges the corridor test of Eq. (1). Randomized smoothing bounds the *stochastic* part, and its claim concerns the model's decision at a point. PAC-Bayes bounds the gap between the risk this model showed on its training mission mix and its risk on this sortie: a statement about the model, constant across the flight. MWU regret bounds the defender's cumulative *policy-switching* loss against an adapting jammer, a statement about the time axis that is vacuous at any single instant.

Because a trajectory, a decision, a model, and a policy sequence are distinct objects, the four compose by conjunction rather than selection: a deployment reads its floor as the pointwise minimum $\min\{f_{\text{G}}, f_{\text{RS}}, f_{\text{PB}}, f_{\text{MWU}}\}$, and the binding term identifies which assumption is closest to failing. In the regime this paper evaluates (a bounded time-delay disturbance, in-distribution missions, a non-adaptive adversary) the Grönwall term binds and the other three are slack, which is why the headline result of §X is a Grönwall curve. We nonetheless carry all four, since relaxing any one of those conditions moves the binding term. We are correspondingly explicit about maturity: the Grönwall and randomized-smoothing floors are instantiated numerically on the reference model, the MWU bound at $|S|=4$, and the PAC-Bayes bound is reported as a form awaiting its numeric KL rather than as a number (Table IX).

X. Instantiation and Evaluation

The theorem is only worth as much as the numbers put into it. This section supplies them and then asks three questions of the result. Does the guarantee hold at a detector setting anyone would actually deploy (§X-D)? Does composing the two layers beat either layer alone (§X-E)? And how much of the whole construction rests on measurement rather than on modelling (§X-H)?

A. The Navigation Anchor

The autonomy layer supplies the result for which the whole construction is based. Fig. 5 shows MCR versus J/S for four defence configurations, computed from the released benchmark's 93,600-flight per-flight table. The undefended baseline collapses below the 0.90 floor by $J/S \approx 8$ dB, whereas the strongest configuration holds it to $J/S \approx 25$ dB. For our purpose, the relevant observation is not which defence wins but that the autonomy layer produces a *graded, measurable* outcome a network event can be paired against.

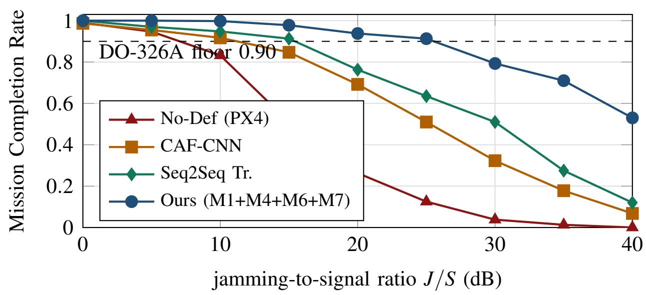

Fig. 5: The navigation anchor: MCR vs. J/S on UAV-EW-BENCH-2026, computed from the released 93,600-flight table (Wilson 95% intervals in the artifact). The graded outcome is what a network attack window is paired against.

B. Reference Constants

Every certified quantity in this section uses the *measured local* Lipschitz constant, not the global power-iteration estimate. The distinction is not cosmetic: the local value is the larger of the two, so it widens the trajectory tube and narrows the certified window, and quoting the global one would flatter every bound reported here. It also moves the tightest corridor across the threshold: at m=2 m the tube is 2.22 m, so θ=0.25 s is certified for $m \geq 5$ m but not at 2 m.

We instantiate the certificate on the reference navigation model, a continuous-time graph model of the GNSS constellation and receiver. Global power-iteration gives a Lipschitz estimate $\hat{L} \approx 1.01$; measuring the constant *along the hidden-state trajectories the integrator actually visits on operating-region inputs* gives a larger local value $\hat{L}_{\text{loc}} \approx 1.18$. We report the tighter local radius as the defensible number: with the certification horizon $T_c = 1$ s and an output tolerance $\varepsilon_{\text{out}} = 0.5$, $R = \varepsilon_{\text{out}} e^{-\hat{L}_{\text{loc}} T_c} \approx 0.153$ (the global estimate gives 0.182). Note the local radius is *smaller*: the bound is not loose in the direction that would flatter us, and the certified regime is governed by real sensitivity rather than an optimistic estimate. The smoothing complement gives a certified ℓ_2 radius ≈ 0.44 at $\sigma = 0.25$ ($\alpha = 10^{-3}$, $n = 200$; measured mean 0.446 on the trained checkpoint), and the MWU defender mixes four policies over N rounds with regret $R(N) \leq \sqrt{N \ln |S|}$, $|S| = 4$. Constants come from the trained reference implementation on a transparent synthetic corpus and are held fixed; Table IX lists each with its provenance tag.

C. A Worked End-to-End Example

To make Eq. (4) concrete we trace one number through it. Take the grid topology, whose routes carry $H = 2$ attacker-controlled hops, and the detector setting $\theta = 0.25$ s. *(1) Threshold to budget.* Since $\theta < s = 5$ s, Lemma 1 gives $\Delta(\theta) \leq H\theta = 0.50$ s of undetected staleness. *(2) Budget to perturbation.* With the measured EKF rate $\gamma = 1.37$ m/s, Eq. (7) gives $\|\delta u\| \leq 0.69$ m; the kinematic bound would instead give 7.5 m. *(3) Perturbation to tube.* With $\hat{L}_{\text{loc}} = 1.18$ and $T_c = 1$ s, the tube radius is $\rho = 0.69\,e^{1.18} \approx 2.25$ m. *(4) Tube to floor.* For any corridor margin $m \geq 2.25$ m the mission is certified to complete; across the margin family $\{2, 5, 10, 20\}$ m the certified floor is positive for all but the tightest 2 m corridor. Under the kinematic bound, step (3) yields $\rho \approx 24.5$ m, exceeding every margin, which is exactly why the empirical calibration, not the theorem, decides the framing. Fig. 6 shows the measured undetected budget feeding step (2), confirming the saturation predicted by Lemma 1. Step (5), the temporal predicate of Eq. (1), is satisfied by a wide margin at this operating point: the induced schedule slip is $\Delta(\theta) \leq 0.50$ s, which is under a 10% tolerance for any mission longer than five seconds.

TABLE IX: Instantiated constants and their provenance. The one pending numeric constant is the PAC–Bayes prior/posterior KL, which we flag rather than substitute.

Component	**Constant**	**Provenance**
Lipschitz–Grönwall	$\hat{L}_{\text{loc}}$=1.181 ⇒ R=0.153; global $\hat{L}$=1.013 ⇒ R=0.182	trained ckpt [fixture]
Rand. smoothing	σ=0.25, R_{ℓ_2}=0.44 (measured 0.446)	trained ckpt [fixture]
PAC–Bayes	McAllester on real R_{emp}; numeric KL pending	methods ch. [pending]
MWU regret	$R(T) \leq \sqrt{T \ln 4}$; T=100 ⇒ 11.77	exact
Unit bridge (RF classes)	`caf_shift_v2`: 0.45 m (Grönwall) / 1.15 m (RS)	synthetic IQ [TEXBAT pending]
$\theta \mapsto \delta$ (delay class)	γ=1.20–1.37 m/s measured; kinematic 15 m/s fallback	real 3 flights [hover]

Why these corridor margins. The family $\{2, 5, 10, 20\}$ m brackets the lateral buffers that operational approval already requires (§II), which is what makes the certified window comparable to a regulatory quantity rather than to a modelling choice. Against the SORA contingency budget [4], [45], 2 m is *tighter* than any compliant lateral buffer and serves as a deliberately adversarial stress case; 5 m sits inside the default static budget; 10 m is representative of a small multirotor's full contingency extent once reaction distance is included; and 20 m covers a fixed-wing platform, whose larger turn radius and cruise speed inflate the manoeuvre term. Reading Fig. 8 against these values converts the abstract question of whether θ is certifiable into the operational one an approval submission must answer: whether the detector setting keeps the worst-case evading tube inside the contingency volume the operator has already declared.

D. The Certified Floor and Its Sensitivity to the Interface Mapping

Fig. 7 reports the certified floor as a function of the detector threshold. It is plotted for two interface mappings rather than one, because the choice between them changes the conclusion. Under the kinematic worst case, $\theta = 0.25$ s falls just outside the certified window; under the empirically measured mapping it falls *inside* at every corridor margin. Fig. 8 quantifies how much room there is: the mapping would have to steepen past $\gamma_{\text{req}} = 6.14$ m/s (at a 10 m corridor) to push the operating point

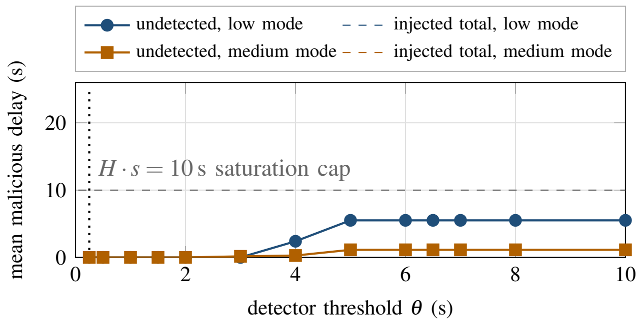

Fig. 6: The measured residual budget behind Lemma 1 (scenario 1, mean over 8 seeds $\times$ 4 policies). The *undetected* delay an evader slips past the detector (solid) saturates well below the *total* injected delay (dashed): beyond the contact-window slack the extra delay produces missed windows flagged at any θ, so the curve the certificate consumes flattens rather than growing.

back out, which is 4.5 times the rate measured under the EKF calibration and 5.1 times the receiver one. This is the paper's most consequential finding for the field: what determines whether cross-layer certification is useful is a *measurement* of how attacks propagate into state error, not the proof machinery.

Would cruise flight cross that threshold? Our calibration flights are low-speed hover and loiter (§XII), so the natural objection is that γ scales with airspeed. An error-budget argument suggests otherwise, and we give it as reasoning rather than measurement. The quantity γ is not the speed at which the aircraft travels but the rate at which its *estimate* of position degrades while a correction is withheld, and during that interval the filter does not freeze: it dead-reckons on inertial measurements, so error accrues at the rate inertial propagation diverges from truth. This is why the measured $\gamma \approx 1.2$–1.4 m/s sits an order of magnitude below the kinematic $v_{\max} = 15$ m/s, which assumes the estimate does not propagate at all. Decomposing that divergence, the one airspeed-scaling term is the cross-track error from a heading error ψ_{err}, growing at $v\,\psi_{\text{err}}$; the remainder (accelerometer bias, attitude error acting on specific force) is set by sensor quality and is largest under the continuous lateral acceleration of a loiter turn, not in straight-and-level cruise. Taking the hover value as the airspeed-independent floor, crossing γ_{req} at $v = 15$ m/s would require $\psi_{\text{err}} \gtrsim (6.14 - 1.37)/15 \approx 0.32$ rad, a sustained heading error near 18°. A PX4-class EKF holds heading uncertainty to a few degrees while GNSS is healthy and raises a yaw-innovation fault well before that, so the regime in which cruise would invalidate the window is one in which the estimator has already declared itself untrustworthy. We therefore expect the window to survive cruise, while noting that only the cruise-inclusive campaign of §XII can settle it.

E. Dominance over Single-Layer Baselines

We state the dominance claim precisely rather than as blanket range-coverage. The composed guarantee dominates

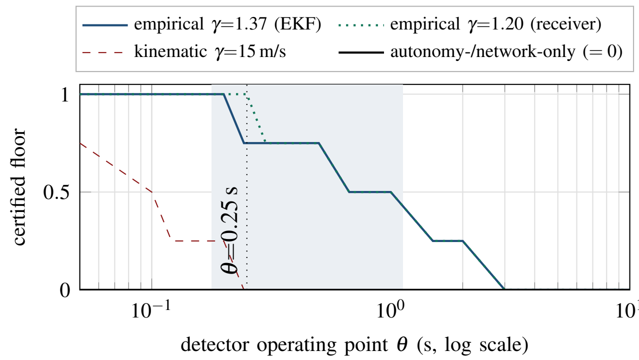

Fig. 7: The headline result: certified floor vs. detector operating point θ for the time-delay class (grid topology, Grönwall tube at the measured local L=1.181, H=2, corridor-margin family $\{2, 5, 10, 20\}$ m). Under the kinematic worst case (red) θ=0.25 s is just outside the certified window; under the empirically measured mapping from real GPS-spoof/jam flights (γ≈1.2–1.4 m/s) it is *inside* at every margin of 5 m or wider (shaded: the EKF window $[0.178, 1.124]$ s at m=10 m); at the tightest 2 m corridor the tube marginally exceeds the margin (2.22 against 2.00 m). Both single-layer baselines certify a zero floor for an evading attacker (black).

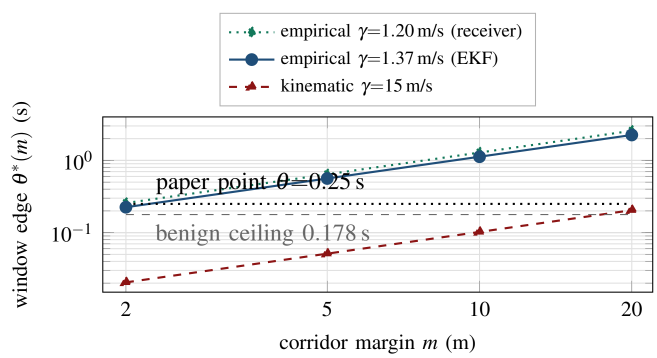

Fig. 8: The decisive sensitivity: the loosest detector setting $\theta^*(m) = m/(\gamma e^{LT_c} H)$ whose worst-case evading tube still fits a corridor of margin m, per interface mapping. A mapping certifies the paper operating point iff its curve lies above the θ=0.25 s line; the feasible region is bounded below by the measured benign residual ceiling (0.178 s, the FPR=0 floor). The measured rates clear the requirement γ_{req}=6.14 m/s at m=10 m by ∼4.5×; the kinematic worst case fails at every margin.

two baselines: (i) *autonomy-only*, a navigation certificate with no detector, which must assume the full unmitigated attack including the missed-window penalties (measured cumulative delay up to 424 s) and therefore certifies a zero floor against an evading attacker; and (ii) *network-only*, a detector with no navigation guarantee, for which any evading attack leaves the aircraft undefended, floor zero by definition. The composed guarantee is the only configuration with a nonzero certified floor at all. Quantifying per seed with the Wilcoxon signedrank test and Holm correction, the composed floor exceeds the autonomy-only floor at all 13 operating points on the θ grid, every adjusted $p < 10^{-20}$. The composition wins because it alone uses the detector to *shrink* the perturbation the certificate must absorb and, via the tightened budget of Lemma 1, the floor saturates rather than collapsing past the knee.

The honest scope: the certified operating window is nonempty but bounded. Under the empirically calibrated interface it is $[0.178, 1.12]$ s at a 10 m corridor, with the operating point $\theta = 0.25$ s comfortably inside; under the conservative kinematic mapping it shrinks to $[0.178, 0.243]$ s and $\theta = 0.25$ s falls just outside. We claim the former and show the latter as the worst-case fallback; we do not claim coverage across the entire relaxation range.

What this comparison does and does not establish: The two baselines are *ablations of our own construction*: they are what remains when one of the two layers is removed. Comparing against them establishes that composition is doing the work, and that the result is not an artifact of either layer alone. It does *not* establish superiority over the alternative approaches to assuring an autonomous system, and we do not claim that it does.

We deliberately do not benchmark against reachability-based verification [25], [26], secure state estimation [28], [29], runtime assurance architectures [31]–[33], or certified reinforcement learning [23], [24], for a reason that is a limitation and not a convenience: those methods answer different questions and none of them consumes a detector operating point, so there is no common axis on which to place them. Reachability bounds the states a system can enter given an input set, but takes the input set as given, which is precisely the quantity our network side supplies. Secure state estimation bounds error given *which* sensors are compromised, rather than modelling a threshold that determines what an adversary may inject. Runtime assurance guarantees safety by switching controllers when a monitor fires, and our certified window is a candidate specification for exactly that monitor rather than a competitor to it.

A fair comparison would require either lifting those methods to consume a detector threshold, or lowering ours to a fixed perturbation ball and comparing tube tightness. The second is feasible and worth doing, and would answer the question "is our Grönwall tube competitive with a modern reachability tool at equal input radius?". We have not done it, and until it is done the contribution should be read as *connecting* two layers that were previously unconnected, not as outperforming the state of the art within either.

F. Certified versus Empirical Floors

The certificate is sound iff the certified floor never exceeds the realised MCR (Proposition 1). At the anchor $J/S = 20$ dB the strongest defence attains empirical MCR $= 0.938$ over 93,600 real benchmark flights, comfortably above the certified target 0.80; the gap is the conservativeness of the worst-case Grönwall bound, and it narrows where the empirically tightened interface replaces the kinematic worst case.

TABLE X: A representative cross-layer pairing: a detector-evading time-delay attack still induces measurable navigation staleness. Values from the released artifact (harness simulation; three-flight δ calibration; benchmark MCR).

Quantity	**Value at representative θ**
Detector threshold θ	0.25 s (paper operating point)
Detector verdict on tuned attacker	evaded (by construction)
Malicious hops H	2 (grid/ring/double-ring routes)
Residual budget $\Delta(\theta) \le H \min(\theta, s)$	≤ 0.50 s (s=5 s; measured cap 4.84 s)
Measured undetected delay (192 runs)	median 0.00 s, max 0.00 s
Staleness, empirical γ=1.37 m/s	≤ 0.69 m (Whelan, hover)
Staleness, kinematic $v_{\max}$=15 m/s	≤ 7.5 m (worst case)
Paired autonomy window	UAV-EW-BENCH at J/S=20 dB
Empirical MCR of paired window	0.938 (strongest defence)
`pairing_basis`	synthetic / measured (per record)

We keep two floors distinct throughout, because conflating them would overstate the guarantee. The reference certificate targets a *certified* MCR ≥ 0.80 at the anchor operating point; the empirical benchmark reports the DO-326A 0.90 *operational* floor holding further into the band. These are different quantities, a conservative proof-derived bound versus a measured rate, and we label every reported number accordingly.

G. A Worked Pairing for Each Evidence Class

Table X reports the pairing at a representative operating point. The `pairing_basis` flag is easiest to understand through one example of each class. *Measured, same platform:* the UAV ATTACK DATASET records a real GPS-jamming flight and its measured position error on one airframe; a pairing joining that flight to the corresponding RF event is tagged `measured_same_platform`, and its δ is a measurement, not a model. *Physics model:* a forwarding-attack window carries a residual budget in seconds; coupling it to a navigation window through the stated relation Eq. (7) yields a `physics_model` pairing: a named, versioned physical relation a reviewer can inspect and challenge, and the class the certificate consumes. *Synthetic alignment:* when a network window from one simulated source and an autonomy window from another are aligned by construction, the pairing is tagged `synthetic_alignment`: maximal coverage, weakest causal claim, and the flag says so. A user needing a strong claim filters to the first class; a user needing breadth accepts the third with its provenance explicit.

H. Provenance Composition of the Corpus

The flag is only useful if its distribution is published, so we report it rather than describe it. Table XI gives the acquisition split across the whole corpus: 200,610 of 431,773 ingested events (46.5%) were captured on real hardware, the remainder generated by simulators of three different kinds. That split is

TABLE XI: Acquisition provenance across the ingested corpus. Counts are from the schema-validated adapter outputs; the split is machine-generated by `experiments/provenance_ledger.py` so the paper cannot drift from the artifact. Acquisition is a property of the *source*; the strictly stronger `pairing_basis` distribution is discussed in the text.

Source	**Acquisition**	**Events**
HCRL_UAVCAN	real testbed	200,000
UAVIDS_2025	simulation (NS-3)	122,171
UAV_EW_BENCH_2026	simulation (physics-informed)	93,600
DATAMUT_SIM	simulation (event)	15,392
UAV_ATTACK_WHELAN	real testbed	610
UAV_CAS	simulation (digital twin)	—
real testbed	captured on hardware	**200,610**
simulation	generated	**231,163**

favourable, but it is also the *weaker* of the two questions a reviewer should ask, and conflating them would overstate the benchmark.

The stronger question is how many *pairings* rest on measurement, and here the honest answer is more restrictive. A `measured_same_platform` pairing requires one source to have observed both the network event and the flight it degraded on a single airframe. Exactly one source in the corpus can do this, the UAV ATTACK DATASET, and although its three flights supply the delay-to-position rate γ that the certificate consumes (Table VII, a real-corpus measurement obtained by self-referencing each flight to its own pre-attack median), the release does not carry machine-readable attack on/off timestamps. Without them the adapter cannot emit the `AttackWindow` boundaries a pairing record requires, and we decline to infer them. The number of released `measured_same_platform` pairings is therefore currently zero: every released pairing is `physics_model` (the network window coupled to a navigation window through the stated relation Eq. (7), and the class the theorem consumes) or `synthetic_alignment` (aligned by construction under Appendix C, with the error budget Eq. (17)).

We state this plainly because it is what the flag exists to expose, and because it locates the benchmark's binding limitation: not the volume of data, of which there is a great deal, but the scarcity of captures observing both layers at once. The headline guarantee does not depend on closing that gap, since γ is measured on real flights whether or not the window boundaries are machine-readable, but a reader filtering the artifact to `measured_same_platform` will correctly retrieve nothing. Obtaining same-platform captures with published attack intervals, across flight regimes rather than hover alone, is consequently the highest-value extension of the corpus.

XI. DISCUSSION

What the composition buys an operator. It turns a network-security knob into a navigation-safety number. A fleet operator setting a detector's false-positive budget currently reasons in packet statistics, with no way to connect that choice to whether the fleet completes its missions. Theorem 1 supplies the missing translation: each detector setting maps to a certified mission floor, so the operator can instead ask what detector tightness a given corridor requires and read the answer off Fig. 8.

Why the interface, not the theorem, is decisive. One might expect the certificate to be the hard part. Our results say otherwise: the tube is a short consequence of a classical inequality and the Lipschitz constant is measured rather than assumed. What decides whether a useful detector setting is certifiable is γ, and the order-of-magnitude gap between the kinematic bound and the measured rate separates a vacuous certified window from one containing the operating point. Tightening cross-layer guarantees is therefore primarily a *measurement* problem, which is why the benchmark and the theorem belong in one paper.

Generality. Although we instantiate on a time-delay forwarding attack and a GNSS navigation model, the chain is agnostic to both ends: any detector with a monotone operating point and any perturbation-to-state mapping slot into Lemma 1 and Eq. (7), leaving the theorem and the four-certificate cover unchanged. Extending the interface catalogue to jamming-intensity-to-error and to bus-level attacks is direct future work.

XII. LIMITATIONS AND FUTURE WORK

We state limitations plainly; none undercuts the contributions, and each defines a concrete extension.

The interface is calibrated, not characterised. This is the paper's binding limitation and condition (C3) of §I. γ is measured on three flights of one PX4 airframe family, all in low-speed hover or loiter. §VII-C shows it is not constant even there, varying by $6.1\times$ once estimated per sample rather than per flight. The campaign covers attack type, GNSS condition and staleness well and airspeed poorly; two factors it cannot vary at all are *platform* (one airframe) and *flight mode* (hover only), and closing those needs flights rather than analysis. The reported spread is consequently a lower bound on the variability a deployment would meet.

The consequence is quantifiable, which is why we state it rather than hedge. If a campaign over those factors found $\gamma > \gamma_{\text{req}} = 6.14\,\text{m/s}$ at a $10\,\text{m}$ corridor, a further four- to five-fold increase over the maximum we observe, the operating point would leave the certified window and the honest framing would revert to the narrow "certify-small, detect-large" regime the conservative kinematic bound already guarantees. §X gives an error-budget argument that such an increase would need a sustained heading error of roughly 18°, but an argument is not a measurement. Two outcomes are possible and both are useful: either γ proves stable across regimes, which would materially strengthen the result, or it does not, in which case the correct model is $\gamma = \gamma(\text{state}, \text{input}, \text{flight mode}, \text{attack type})$ and the certificate must be evaluated at the supremum over the operational design domain. We regard establishing which of the two holds as the highest-value extension of this work.

The certificate assumes the filter re-anchors. Theorem 1 propagates deviation over T_c and relies on Definition 2, that one uncorrupted state update arrives per window. A delay adversary satisfies this by construction, postponing corrections without suppressing them; an adversary denying updates outright does not, and there the certified radius degrades as e^{LT_c} in the achieved denial duration. Certifying against sustained denial needs either a link-layer bound on outage length or a fallback whose guarantee is independent of L, which is the runtime-assurance direction of §XI. More generally the Grönwall bound is conservative by design, which is why the architecture is a four-certificate cover; fusing the deterministic, smoothing, and PAC-Bayes floors into one tighter envelope is open.

Evidence-class imbalance. Only one of the six datasets couples a network/RF event to a measured flight on one airframe, so it is the sole source of `measured_same_platform` pairings. This reflects what public data exists, and is why the honesty flag is mandatory rather than optional.

Calibration constants that remain data-gated. The unit bridge mapping a raw GNSS position error into receiver-level feature space, which serves the RF-manipulation classes rather than the time-delay class the theorem targets, is calibrated on a synthetic signal corpus pending access to real in-phase and quadrature recordings; and the PAC-Bayes bound Eq. (14) is wired to the trained model's measured empirical risk but awaits its numeric KL divergence. Neither gates the headline result, and both are tracked explicitly so that no synthetic constant is reported as real.

Analytical navigation anchor and detector scope. Completion outcomes come from a physics-informed analytical Monte-Carlo backend calibrated to full-fidelity anchors, so we describe them as a calibrated analytical benchmark. We also instantiate the harness on one forwarding detector whose operating point is a single clean threshold, which makes the cross-layer sweep unambiguous; extending to multi-dimensional operating points requires a higher-dimensional sweep, not a schema change.

Threats to validity. Construct: the honesty flag bounds the risk that a recorded pairing misrepresents the causal link it claims, and a reviewer can restrict any analysis to the measured class. *Internal:* two of our own measurement errors were caught because the provenance discipline made them visible, and every figure is regenerated from committed data. *External:* four of six sources are simulation or calibrated twin, so absolute detector-performance numbers are corpus-specific; what transfers is the *structure*, a property of store-and-forward scheduling and estimator behaviour rather than of any one simulator.

XIII. Conclusion

Attacks on a UAV's network and degradation of its flight have been approached separately, because nothing available made it possible to approach them together. We built the benchmark that makes the coupling representable and proved, on exactly its objects, the conditional guarantee that the coupling enables: a two-layer schema whose unit is the cross-layer pairing, six validated adapters, and a detector-operating-curve harness on the benchmark side; a residual budget that measurement tightens into the saturating bound $\Delta(\theta) \leq H \min(\theta, s)$, an interface rate calibrated on real spoofing and jamming flights, and a composition theorem chaining a detector's operating point to a certified mission-completion floor on the guarantee side.

Three findings survive scrutiny. Among the configurations compared, the composed guarantee is the only one certifying a non-zero mission floor against a detector-evading attacker, and it does so at every operating point. The empirically calibrated interface, an order of magnitude tighter than the kinematic worst case, brings a deployed detector's operating point inside the certified window, making the guarantee operationally relevant rather than merely formal. And the binding constraint on cross-layer certification is the measurement of how attacks propagate into state error, not the proof machinery: the interface, not the theorem, decides the outcome.

That third finding cuts both ways, and we end on it deliberately. Because the interface decides the outcome, the result is only as strong as the calibration behind it, and ours rests on three flights of one airframe in hover, over which γ already varies by $6.1\times$ with attack type, satellite visibility and staleness. What we offer is therefore a conditional guarantee under an empirically calibrated cross-layer interface: the conditions are stated in §I, the calibration is released, and the experiment that would confirm or refute its stability is specified in §XII. Under those conditions, UAV network security and UAV autonomy security are not two problems but two ends of one certificate.

XIV. Ethics Considerations

This work involves no human subjects and no personally identifying data. All six datasets are public research corpora released for security research; we redistribute none of the raw third-party data, shipping only integration scripts and unified labels so that each source is fetched from its origin under its own licence. No live aircraft, production network, or third-party system was attacked: every attack reported here is either replayed from a published capture or generated in simulation on infrastructure we control.

On dual use, our harness quantifies how an attacker can stay under a detector's threshold. We judge disclosure net-positive: staying under a published threshold is the definitional behaviour of an evading attacker and is already assumed in the detector's own analysis, our central result *bounds* what such an attacker achieves and shows that it saturates, and the practical consequence is a defensive configuration procedure, namely reading the required detector tightness off Fig. 8 for a given corridor margin. We disclose no previously unknown vulnerability in a deployed system, so no coordinated-disclosure process applied; the underlying detector's authors are aware of this analysis of their operating point.

XV. Open Science and Artifact Availability

Everything needed to reproduce this paper is released as an open artifact: the schema, the six ingestion adapters, the operating-curve harness, the certificate engine, the statistical analysis, and the figure generator. Every figure and table in this paper is machine-generated from committed result files rather than transcribed, so a reader can regenerate them from the released data; each carries a provenance tag (real-corpus, simulation-derived, or stated modelling parameter) in an artifact ledger. The harness is deterministic per seed, and headline comparisons are released per seed with their nonparametric tests. Two measurement errors of our own were caught by this discipline before publication: a sweep harness that accumulated across operating points, and a set of bus-attack labels that are not machine-readable per scenario and which we therefore derive and flag rather than guess. We intend to submit the artifact for evaluation. The repository is anonymised for review and will be de-anonymised on acceptance.

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Appendix

The composition proved in this paper spans two research communities that share very few conventions: one reasons about packets and detector thresholds, the other about trajectories and certified radii. This section fixes the single system model both halves of the argument refer to, namely how an autonomous UAV closes its navigation loop, which of its interfaces an adversary can reach, and what a robustness *certificate* does and does not promise, so that the interface mapping of §VII joins two precisely defined quantities rather than two informal ones. Acronyms are defined at first use and not tabulated; Table XII is the single reference for mathematical symbols. A reader fluent in both UAV autonomy and certified robustness may proceed directly to §II.

A. The UAV Navigation Loop

An unmanned aerial vehicle (UAV, colloquially a drone) closes a control loop around a *state estimate*. An onboard computer fuses fixes from a Global Navigation Satellite System (GNSS, of which the American GPS is one constellation) receiver, inertial measurements from accelerometers and gyroscopes (the IMU), and barometric height into a best guess of where the aircraft is and how fast it is moving. On the PX4 autopilot family used throughout this paper the fusion is performed by an extended Kalman filter (EKF), a standard recursive estimator that weighs each sensor by its expected noise [39]. The flight controller steers to keep the estimate on the planned route.

Two properties of this loop drive everything that follows. First, the loop is only as good as its *freshest trustworthy input*: if GNSS is denied, the EKF "coasts" on inertial data and its position error grows with time. Second, the loop is *networked*. Swarm members and the ground control station exchange corrections and commands over radio links: a command-and-control (C2) link, typically speaking the Micro Air Vehicle Link (MAVLink) protocol, a multi-hop mesh in which peers relay each other's traffic during periodic contact windows, forming a delay-tolerant network (DTN), and an intra-vehicle bus, the UAV Controller Area Network (UAVCAN, also DroneCAN), connecting the flight controller to motor controllers and sensors. An attacker on the mesh can therefore reach the navigation loop without ever touching the radio-frequency (RF) front end.

B. GNSS Spoofing and Jamming

Electronic warfare (EW) attacks the sensing physics. A *jammer* raises the jamming-to-signal power ratio (J/S, in decibels) until the receiver loses satellite lock; a *spoofer* broadcasts counterfeit satellite signals so the receiver locks onto a false position [46], [47]. *Network attacks* target the messages. A *time-delay attack*, the class our theorem covers, has a compromised relay hold packets before forwarding them, so downstream consumers act on stale state [2], [3]. Neither needs to break cryptography; both degrade the control loop through legitimate-looking inputs, which is precisely why detection and robustness must be composed rather than assumed.

C. The Scope of a Robustness Certificate

A *certified* robustness statement differs from an empirical evaluation: instead of reporting that attacks in a test set were survived, it proves that *every* perturbation within an explicitly stated budget is survived. The price is conservatism: certified bounds are worst-case, so a certified floor typically sits below what is observed empirically. We keep the two kinds of number separate and label them throughout: *certified* floors come from proofs, *empirical* rates from measured flights. We use two flavours of certificate: deterministic Lipschitz-type bounds (if a model cannot amplify input changes by more than a factor L, a bounded input change yields a bounded output change) and probabilistic randomized-smoothing bounds (adding calibrated

TABLE XII: Mathematical notation. Every symbol is defined where first used; this table is the single reference.

W_n, W_a	network / autonomy attack window
P	cross-layer pairing tuple, Eq. (3)
$x(t) \in \mathbb{R}^n$	navigation state (position, velocity, ...) at time t
$u(t)$, δu	sensor input to the navigation dynamics; its perturbation
F	drift (right-hand side) of the navigation dynamics Eq. (10)
T, κ	nominal mission duration (s); schedule tolerance as a fraction of T
T_c	certification horizon: interval between uncorrupted state updates over which the trajectory tube is propagated (1 s)
m	safe-corridor margin: allowed lateral deviation, metres
D_θ, θ (ε)	network detector; its operating point (tolerated per-hop residual delay, s)
s, P_c	contact-window slack (5 s); contact period (60 s)
H	number of attacker-controlled (malicious) hops on a route
$\Delta(\theta)$	residual budget: worst undetected cumulative delay, s
$\delta(\theta)$	induced perturbation on the navigation input
γ	delay-to-position rate: metres of error per second of degradation
$v_{\max}$	kinematic worst-case speed bound (15 m/s)
L, $\hat{L}$, $\hat{L}_{\text{loc}}$	(estimated / local) Lipschitz constant of F
$\rho(\theta)$	trajectory-tube radius $\delta(\theta)e^{LT_c}$, metres
$\theta^*(m)$	loosest θ whose tube fits margin m (window edge)
$f(\cdot)$	certified MCR floor as a function of the perturbation budget
R, σ, α, n	certified input radius; smoothing noise, confidence, samples
R_{emp}, KL	empirical risk and prior-posterior divergence, Eq. (14)
$\lvert S\rvert$	size of the defender's policy set in the MWU certificate (4)

noise and taking a majority vote yields a radius within which the decision provably cannot flip) [17]. The regulatory anchor for "mission survived" is the airworthiness-security process standard DO-326A/ED-202A [5], whose industry-acceptable completion floor of 0.90 appears in our figures as a horizontal reference line.

We instantiate the harness on a threshold detector because its operating point is a single, physically meaningful scalar, which makes the cross-layer sweep unambiguous. Since most deployed UAV intrusion detection uses learned classifiers, it is worth showing that the framework does not depend on that choice.

A learned detector emits a score $\sigma(a) \in [0,1]$ and flags an event when $\sigma(a) \geq P_{\text{th}}$; sweeping P_{th} traces the familiar receiver operating characteristic, each cutoff being a (FPR, TPR) pair exactly as each ε is in Fig. 3. The only object the composition consumes is the residual budget (the largest attack effect that survives the detector), so the framework generalises as soon as P_{th} maps to one, which it does through the detector's *score-to-effect* profile. Let $g(\cdot)$ be the monotone-increasing map from an attack's physical magnitude μ (here, injected per-hop delay in seconds) to the detector's expected score, $\sigma = g(\mu)$, measured by running the detector over labelled attacks of known magnitude, which every corpus in §V supports. An attacker evading the cutoff must keep $g(\mu) < P_{\text{th}}$, hence

$$\mu \;<\; g^{-1}(P_{\text{th}}) \;=:\; \theta_{\text{eff}}(P_{\text{th}}), \tag{15}$$

and Lemma 1 applies verbatim with θ replaced by the *effective* operating point θ_{eff}. The threshold detector is simply the special case $g = \text{id}$, for which $\theta_{\text{eff}} = \varepsilon$. Two properties of g carry over into the guarantee. Monotonicity is what makes g^{-1} well defined and the certified window an interval rather than a union; where a classifier is non-monotone in μ, one takes the conservative $\theta_{\text{eff}} = \sup\{\mu : g(\mu) < P_{\text{th}}\}$, which is sound but looser. And because a learned score is stochastic, g is a distribution rather than a function, so θ_{eff} must be read at an upper confidence bound on the score, which is precisely the regime in which the randomized-smoothing certificate of §IX, rather than the deterministic Grönwall tube, is the appropriate downstream bound.

Any detector exposing a tunable cutoff and a measurable score-to-magnitude profile therefore slots in without a schema change: the harness sweeps P_{th} instead of ε, records $(\text{FPR}, \text{TPR}, \theta_{\text{eff}})$ per cutoff, and the downstream chain is unchanged. What the framework requires is not that a detector be deterministic, but that its operating point reduce to a bound on the physical attack magnitude that survives it.

Because the sources share no global clock, a `synthetic_alignment` pairing must state exactly how a network window and an autonomy window from different captures are placed on a common axis, and how much error that placement can introduce. Left implicit, this is the step at which a cross-layer benchmark could silently manufacture, or destroy, the very effect it reports.

Let the network source have relative clock $\tau_n \in [0, T_n]$ and the autonomy source $\tau_a \in [0, T_a]$, each measured from its own capture start. An alignment is an affine map

$$\phi(\tau_n) \;=\; a\,\tau_n + b, \qquad a = \frac{T_a}{T_n}, \tag{16}$$

whose scale a normalises the two capture durations and whose offset b is fixed by *anchoring attack onsets*: we set b so that $\phi(t_0^n) = t_0^a$, where t_0^n is the first labelled malicious event in W_n and t_0^a the onset of degradation in W_a. Onset is not eyeballed: t_0^a is the first sample at which the autonomy metric exceeds its pre-attack median by more than $3\hat{\sigma}$, with $\hat{\sigma}$ the robust (median-absolute-deviation) scale of the pre-attack segment. Both anchors are therefore defined by the source labels and the source's own noise floor, not by the analyst.

Three error terms bound the residual misalignment e_{align}:

$$e_{\text{align}} \;\leq\; \underbrace{\tfrac{1}{2}(\Delta t_n + \Delta t_a)}_{\text{quantisation}} + \underbrace{|a-1|\,T_a}_{\text{rate mismatch}} + \underbrace{e_{\text{onset}}}_{\text{anchor detection}}, \tag{17}$$

where $\Delta t_n, \Delta t_a$ are the two sources' sampling intervals (a per-flow network record and a 50 Hz telemetry stream give $\frac{1}{2}(\Delta t_n + \Delta t_a) \leq \frac{1}{2}(1.0 + 0.02) = 0.51$ s), the rate term vanishes when both captures are replayed at their native rate ($a = 1$, the case for every pairing we release), and $e_{\text{onset}} \leq \Delta t_a$ is the one-sample latency of the $3\hat{\sigma}$ crossing. For the released `synthetic_alignment` pairings this gives $e_{\text{align}} \leq 0.53\,\text{s}$.

This bound is what keeps the alignment from biasing the reported staleness. The quantity the certificate consumes, $\Delta(\theta)$, is computed *entirely within the network source* from per-hop residual delays; the alignment determines only which autonomy window a given network window is paired *with*, never the numeric value of $\Delta(\theta)$. Misalignment can therefore attach a pairing to the wrong autonomy interval, but it cannot inflate or deflate $\Delta(\theta)$ itself. Its effect is confined to the empirical MCR column, and it is bounded there too: since $e_{\text{align}} \leq 0.53\,\text{s}$ is two orders of magnitude below the $60\,\text{s}$ contact period that separates distinct attack windows, no released pairing can be displaced onto a neighbouring window. Pairings whose alignment error would exceed one tenth of the shorter window's duration are rejected by the adapter rather than emitted with a caveat.

Proof of Theorem 1. Fix a mission and let $x_{\text{nom}}(t)$ be the nominal trajectory (the solution of Eq. (10) under the attack-free input $u(t)$) and $x_{\text{att}}(t)$ the trajectory under $u(t)+\delta u(t)$, both started from $x_{\text{nom}}(0) = x_{\text{att}}(0)$. Write $e(t) = x_{\text{att}}(t) - x_{\text{nom}}(t)$. From Eq. (10), $\dot{e}(t) = F(x_{\text{att}}, u+\delta u) - F(x_{\text{nom}}, u)$, so L-Lipschitzness gives $\|\dot{e}(t)\| \leq L\|e(t)\| + L\|\delta u(t)\|$. Integrating with $e(0) = 0$,

$$\|e(t)\| \;\leq\; L\int_0^t \|e(\tau)\|\,d\tau \;+\; L\int_0^t \|\delta u(\tau)\|\,d\tau.$$

Applying the integral form of Grönwall's inequality [40] to the non-decreasing forcing term $g(t) = L\int_0^t \|\delta u\|\,d\tau$ yields $\|e(t)\| \leq g(t)e^{Lt}$, and bounding the input by $\|\delta u(\tau)\| \leq \delta(\theta)$ gives, at the certification horizon,

$$\|e(T_c)\| \;\leq\; \delta(\theta)\big(e^{LT_c} - 1\big) \;\leq\; \delta(\theta)e^{LT_c}, \tag{18}$$

which is Eq. (11); we report the closed-form right-hand bound. By Definition 2 the deviation is reset at least once per window of length T_c, so the same bound holds on each window and therefore on the supremum over $[0,T]$. Now suppose $\rho(\theta) \leq m$. The nominal trajectory lies in the safe corridor by construction, and every point of the attacked trajectory is within $\rho(\theta) \leq m$ of it by Eq. (18); hence the attacked trajectory never exits the corridor over $[0,T]$, discharging the spatial predicate of Eq. (1). For the temporal predicate, the only mechanism by which a D_θ-evading time-delay attack displaces arrival is the undetected forwarding delay it accumulates, which Lemma 1 bounds by $\Delta(\theta)$; hence $t_{\text{arr}} \leq T + \Delta(\theta)$, and $\Delta(\theta) \leq \kappa T$ gives $t_{\text{arr}} \leq (1+\kappa)T$ as required. Both predicates holding, the mission completes. Taking the probability of $\{\rho(\theta) \leq m\} \cap \{\Delta(\theta) \leq \kappa T\}$ over the mission distribution, the only source of randomness once θ is fixed, since $\delta(\theta)$ and $\Delta(\theta)$ are deterministic bounds, gives Eq. (12). Composing $\theta \mapsto \Delta(\theta)$ (Lemma 1), $\Delta \mapsto \delta$ (§VII), and $\delta \mapsto \text{MCR}$ establishes the chain Eq. (4). □